UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	PROGRAMMING I
COURSE CODE	:	CSC402
EXAMINATION	:	NOV 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC402 mock paper (November 2026). Generated answers show working; all questions are deterministic and paper-derived; no lecture slides are used.

2. Marking points show the expected answer and the allocation of marks; unresolved slots would remain visible.

3. Part A = 10 x 1 mark; Part B = 5 questions (60 marks); Part C = 2 x 15 marks; total = 100 marks.

4. Questions: 17. Slots: 22. Placeholders: 0. No LLM call. Questions and answers are for local study use only.

5. Structure derived from 21 unique CSC402 past papers; no past-paper text is copied.

FOR MARKERS' USE ONLY

This marking scheme consists of 6 printed pages

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PART A — ANSWER KEY

Q	Ans	Question	Solution
A1	D	Which statement immediately exits a loop?	break terminates the nearest loop or switch.
A2	D	Which declaration creates a floating-point variable named price?	float is used for real numbers.
A3	B	What does a compiler do?	A compiler translates the complete source file into object code.
A4	B	Which identifier is valid in C++?	An identifier may contain letters, digits and underscores, but cannot start with a digit or be a keyword.
A5	C	Which declaration creates an integer array of five elements?	int a[5]; reserves five integer elements.
A6	A	Which statement is most suitable for selecting among several constant cases?	switch selects a case based on a constant expression.
A7	B	Which part of a for loop is executed after each iteration?	The update expression runs after each iteration.
A8	C	Which phase of the program development life cycle defines what the program must do?	Analysis establishes the requirements before design and coding.
A9	A	Which keyword prevents a variable's value from being changed after initialisation?	const defines a read-only variable.
A10	D	Which operator sends data to cout?	The insertion operator << writes to an output stream.

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PART B — WORKED ANSWERS

QUESTION 1

Topic: debugging testing

a) Define algorithm and pseudocode.

An algorithm is a finite sequence of precise steps that solves a problem. Pseudocode is a structured English-like description of an algorithm.

Marking: Award 1 mark for each correct definition.

(2 marks)

b) Explain THREE types of programming errors and give one example of each.

Syntax error: missing semicolon. Runtime error: division by zero. Logic error: using < instead of <= in a condition.

Marking: Award 0.5 mark for each type and 0.5 mark for a valid example.

(3 marks)

QUESTION 2

Topic: fundamentals

a) Draw a flowchart to find the largest of three values.

Start -> read a, b, c -> largest = a -> if b > largest then largest = b -> if c > largest then largest = c -> display largest -> end.

Marking: Award marks for input, initial assumption, two comparisons, output and flow.

(8 marks)

b) Write pseudocode and a complete C++ program for the flowchart.

Use if statements to update largest, then cout the result.

Marking: Award marks for correct comparisons, variable updates and output.

(5 marks)

QUESTION 3

Topic: data types

a) Explain the difference between implicit and explicit type conversion, and give one C++ example of each.

Implicit conversion is automatic (int to double); explicit conversion uses a cast such as double(total) or static_cast<double>(total).

Marking: Award marks for both definitions and valid examples.

(6 marks)

b) Write a C++ program segment that calculates the average of three integer marks as a double and displays it to two decimal places.

double average = (m1 + m2 + m3) / 3.0; cout << fixed << setprecision(2) << average;

Marking: Award marks for floating-point division, fixed/setprecision and output.

(5 marks)

QUESTION 4

Topic: control structures

a) Trace the program and show the exact output.

1\n12\n123

Marking: Award marks for each line and the exact row lengths.

(6 marks)

b) Identify TWO logic errors in a version that prints four rows but reuses j without resetting it. Rewrite the corrected code.

The inner loop must initialise j = 1 for every row and the outer loop must run to 4. Correct code: for (int i=1;i<=4;i++){for(int j=1;j<=i;j++) cout<<j; cout<<endl;}

Marking: Award marks for each error and the corrected loop.

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(9 marks)

QUESTION 5

Topic: arrays

a) Declare an array named prices with the values 20.5, 55.0, 15.9, 75.0 and 10.0. Display all values.

double prices[5] = {20.5, 55.0, 15.9, 75.0, 10.0}; for (int i=0;i<5;i++) cout << prices[i] << endl;

Marking: Award marks for correct type, initialiser, loop and output.

(6 marks)

b) Write a program segment that applies a 10% discount to prices above RM50 and stores the results in a second array. Display both arrays.

double sale[5]; for(int i=0;i<5;i++){ sale[i] = prices[i] > 50 ? prices[i]*0.9 : prices[i]; } then display both arrays.

Marking: Award marks for second array, condition, discount, preservation and output.

(10 marks)

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PART C — WORKED ANSWERS

QUESTION 1

A bookstore records up to five books. Each book has a title, price and quantity in stock.

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

```
#include <iostream>
#include <string>
using namespace std;
struct Book {
    string title;
    double price;
    int quantity;
};
int main() {
    Book books[5];
    double totalValue = 0;
    int mostExpensive = 0;
    for (int i = 0; i < 5; i++) {
        cin.ignore();
        cout << "Title: ";
        getline(cin, books[i].title);
        cout << "Price and quantity: ";
        cin >> books[i].price >> books[i].quantity;
        totalValue += books[i].price * books[i].quantity;
        if (books[i].price > books[mostExpensive].price) {
            mostExpensive = i;
        }
    }
    cout << "Total inventory value: RM" << totalValue << endl;
    cout << "Most expensive: " << books[mostExpensive].title
         << " (RM" << books[mostExpensive].price << ")" << endl;
    return 0;
}
```

Marking: Award 3 marks for struct definition, 4 for input, 4 for inventory value, 3 for most expensive and 1 for output.

(15 marks)

QUESTION 2

A cafe sells three items: coffee RM8.50, tea RM6.00 and juice RM9.50. A 10% discount is given when the subtotal is at least RM50.

Write a complete C++ program that reads the item code and quantity, calculates the subtotal and discount, and displays a formatted receipt with the item name, quantity, subtotal, discount and amount to pay.

```
#include <iostream>
#include <iomanip>
#include <string>
using namespace std;
int main() {
    int code, quantity;
    double price = 0, subtotal, discount = 0;
    string name;
    cout << "Item code (1 coffee, 2 tea, 3 juice): ";
    cin >> code;
    cout << "Quantity: ";
    cin >> quantity;
    if (code == 1) { name = "Coffee"; price = 8.50; }
    else if (code == 2) { name = "Tea"; price = 6.00; }
    else if (code == 3) { name = "Juice"; price = 9.50; }
    else { cout << "Invalid item code" << endl; return 1; }
    subtotal = price * quantity;
    if (subtotal >= 50) discount = subtotal * 0.10;
    cout << fixed << setprecision(2);
    cout << "Item: " << name << endl;
    cout << "Quantity: " << quantity << endl;
```

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```
    cout << "Subtotal: RM" << subtotal << endl;
    cout << "Discount: RM" << discount << endl;
    cout << "Amount to pay: RM" << subtotal - discount << endl;
    return 0;
}
```

Marking: Award 3 marks for input, 5 for price/subtotal, 4 for discount, 2 for formatting and 1 for completeness.

(15 marks)

END OF MARKING SCHEME

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